

PORT WELLER, ON, Canada

WMO: 714320

Lat: 43.245N

Lon: 79.218W

Elev: 259

StdP: 14.56

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	12.0	1.4	6.0	10.4	4.9	7.2	14.0	37.0	24.8	35.2	24.4	14.5	250	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	84.8	72.6	82.1	71.6	79.7	70.5	75.6	81.0	73.9	79.0	72.2	77.4	6.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	128.6	79.1	72.3	120.8	77.2	70.5	113.4	75.6	39.4	80.9	37.6	79.1	36.1	77.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.1	28.3	24.9	DB	5.1	89.3	4.9	3.1	1.6	91.5	-1.3	93.4	-4.1	95.1	-7.7	97.4	(4)
(5)			WB	4.0	77.8	5.4	2.5	0.1	79.6	-3.0	81.1	-6.0	82.5	-10.0	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	27.2	28.3	34.7	44.0	54.6	65.4	71.9	71.6	65.6	53.9	42.6	33.7	(6)	
	DBStd	17.37	9.20	8.20	7.78	6.71	7.08	6.34	4.85	4.55	5.99	7.13	7.06	7.63	(7)	
	HDD50	2809	709	607	477	203	28	0	0	0	0	39	240	506	(8)	
	HDD65	6241	1173	1027	941	629	333	73	3	4	64	351	673	970	(9)	
	CDD50	2654	1	0	2	24	171	461	678	670	468	161	17	1	(10)	
	CDD65	609	0	0	0	0	12	84	216	209	81	7	0	0	(11)	
	CDH74	2647	0	0	0	1	57	366	1048	868	290	17	0	0	(12)	
	CDH80	494	0	0	0	0	5	72	224	144	49	0	0	0	(13)	
(14) Wind	WSAvg	10.3	13.6	12.8	12.4	11.1	8.2	7.1	6.8	7.2	8.8	11.0	12.1	13.0	(14)	
(15) Precipitation	PrecAvg	33.6	2.7	2.2	2.4	3.0	3.0	3.1	3.1	3.0	3.3	3.1	2.7	2.8	(15)	
	PrecMax	43.5	4.6	4.7	4.2	5.1	6.3	7.0	7.4	5.5	7.5	6.9	4.8	6.1	(16)	
	PrecMin	26.9	1.5	0.9	0.7	1.3	0.8	0.8	0.7	1.3	1.5	1.1	1.1	1.1	(17)	
	PrecStd	4.8	0.9	0.9	1.0	1.1	1.5	1.5	1.6	1.1	1.4	1.4	0.9	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	54.3	60.0	69.4	79.8	86.0	89.2	88.3	85.3	76.4	64.4	57.2	(19)	
		MCWB	52.4	47.7	51.3	58.7	66.2	71.7	75.6	75.9	72.4	66.2	55.9	53.1	(20)	
	2%	DB	47.7	46.7	54.7	63.3	74.6	81.9	85.3	83.6	80.5	71.7	60.5	50.8	(21)	
		MCWB	44.4	42.2	48.5	54.0	63.9	70.1	73.0	72.7	71.0	63.5	54.8	47.4	(22)	
	5%	DB	42.7	41.5	49.4	58.5	70.1	78.1	82.3	80.9	77.1	67.7	56.4	46.7	(23)	
		MCWB	39.8	37.9	44.5	50.9	61.2	68.4	71.6	71.7	69.6	60.9	51.4	43.9	(24)	
	10%	DB	38.3	37.9	44.0	53.6	65.7	74.9	79.5	78.7	74.3	64.1	52.7	43.2	(25)	
		MCWB	36.1	35.1	40.5	47.7	58.8	67.1	70.7	70.8	68.2	59.1	48.6	40.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	49.3	53.7	60.7	68.4	74.6	78.4	78.0	75.1	69.5	59.5	53.4	(27)	
		MCDB	54.2	52.6	59.0	65.7	76.4	80.2	86.1	84.1	80.9	74.6	62.8	57.0	(28)	
	2%	WB	45.0	43.3	48.9	55.7	65.2	71.9	75.9	75.8	73.0	64.7	55.4	47.8	(29)	
		MCDB	47.1	47.1	53.2	61.3	73.0	78.4	81.7	80.6	77.8	70.2	59.2	50.3	(30)	
	5%	WB	39.7	38.7	44.7	51.7	62.6	69.9	74.0	73.8	71.1	62.0	51.9	44.2	(31)	
		MCDB	42.3	41.5	49.1	57.5	69.3	76.5	79.4	78.7	75.6	66.1	56.1	46.3	(32)	
	10%	WB	36.5	35.6	40.7	47.7	59.8	67.9	72.2	72.0	69.2	59.9	49.0	41.0	(33)	
		MCDB	38.5	37.7	44.1	52.8	65.2	73.6	77.4	76.8	73.6	63.8	52.7	43.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	9.9	10.3	11.5	13.2	12.4	11.9	11.0	10.8	10.0	9.0	8.2	(35)	
		MCDBR	14.2	15.7	18.0	21.2	20.6	17.3	15.4	14.2	14.9	15.2	15.1	12.8	(36)	
	5% WB	MCWBR	12.7	13.1	13.2	13.7	11.7	9.4	7.7	7.4	8.1	9.5	11.4	10.8	(37)	
		MCWBR	13.3	15.8	17.6	20.4	19.5	16.1	13.8	12.5	12.6	13.1	14.2	12.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.300	0.327	0.341	0.383	0.413	0.443	0.440	0.425	0.390	0.368	0.319	0.304	(40)	
		taud	2.429	2.343	2.388	2.307	2.272	2.224	2.254	2.285	2.360	2.391	2.496	2.461	(41)	
	Ebn at Noon		265	274	285	280	273	265	264	264	265	256	256	252	(42)	
		Edh at Noon	24	31	33	39	42	44	42	40	34	29	23	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		430	672	1060	1403	1717	1859	1867	1652	1318	796	496	347	(44)	
	RadStd		39	59	112	149	181	137	141	86	118	72	66	41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page